

An Analytical Approach to What Makes A Song Popular and How This Has Changed Over Time

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Abstract

The songs we listen to as a society are a significant part of the popular culture, and looking at the songs that were popular at any given point in time can give indications as to the zeitgeist of the location and period. In modern times, there is a large music and recording industry dedicated to producing songs that they hope will become popular and dominate the charts for long periods of time. Due to the significant cultural and financial impact of producing popular songs, it is therefore worth understanding the features of songs that can contribute to their popularity. In this project, we find the features of songs that influence their popularity, analyze how these features have changed over time, and attempt to classify songs as hit songs based on their features.

1 Introduction

1.1 Motivation

Studying how music tastes have evolved is interesting because it can reflect on how society as a whole has evolved. Tastes in music are often shaped by the culture of the society, and new genres of music can come from that same culture. Even down to a specific person, taste in music is often guided by brain region activations, dopamine levels, and association with positive mood and affect [8].

Studies have also shown that there are significantly more songs, and artists populating the top charts than before [2]. There are continued efforts to show whether this is indicative of a cultural convergence or divergence. While this project does not make conclusions as to the nature of culture shifts, it does attempt to reduce the songs down to their most important features and study them from this perspective.

The music industry has seen a rise in the use of big data and analytics to study songs from a more quantitative perspective as well. From a music streaming perspective, large online music websites, such as Spotify, Apple Music, Amazon Music, Tencent, and YouTube all have access to user data and music consumption data. These companies all make efforts to analyze users' listening habits and relate that to song listening worldwide to reasonably predict what songs a user might want to listen to and recommend them to the users with recommendation algorithms. From a recording perspective, record companies can analyze trends in what songs become popular, and in turn, will encourage their artists to make music that they believe will exploit the patterns that today's popular music exhibits. Collectively, billions of

dollars are invested in finding new musical talent [7]. The importance of data analytics in song popularity (i.e. "hit song science") is the primary motivator for this project.

1.2 Key Contributions

In summary, this project makes the following contributions:

1. An overview of audio features of songs and a general understanding of how they are calculated.
2. An analysis of song features and how they have evolved over time.
3. A clustering of songs in an attempt to identify types of songs that become popular.
4. An attempt to predict the popularity of songs based on their features.
5. An association of the key features in the prediction of a song's popularity and the features that have notably changed over time.

2 Background and Related Work

2.1 What Can Influence a Person To Like a Song?

Even with the vast range of songs that are published today, there have been studies from a neurological perspective and by music theorists into what traits will lead to songs being enjoyable. In general, humans tend to like songs that exhibit familiar patterns and attributes. Music producers will exploit this by utilizing these patterns, mixed in with some variations to keep us mostly in the realm of what is expected, but always waiting for some sort of surprise. For instance, songs can use "deceptive cadence", repeating patterns to lull us to expect only those patterns for the duration of the song, and then abruptly switching to catch us off guard [6], [8]. The patterns of each song we listen to will leave an imprint in our brain, which will influence how we perceive future songs, and how we associate songs with certain events. This is why even decades later, we can have positive associations with songs we heard in our childhood.

2.2 How Can Features of Songs Be Extracted?

In order to conduct meaningful analysis of the audio features of a track, we need to be able to derive those features from the data of the track itself. While the focus of this project is an analysis of high-level "meta-features" that are derived from lower level analysis of the raw musical data, it is worth a quick overview of how the features of a track would be extracted in the first place [1].

Starting with a raw audio track file, key components of the track data that are typically considered include:

- **Time Period:** How long a certain sound lasts, i.e. how many seconds it takes to complete one cycle of vibrations.
- **Amplitude:** the sound intensity, measured in decibels (dB). People perceive this as the 'loudness' of a song.
- **Frequency:** indicator of how many sound vibrations happen per second, measured in Hertz (Hz). People perceive this as the pitch of a sound.

Music stored in Audio Interchange File Format (.aiff) or Waveform Audio File Format (.wav) files is typically used to extract features as these file formats do not compress the original sound recordings. These files then are transformed into visual representations of the audio files that can be more readily understood by ML models that produce the features. The visual representations can include:

- **Waveform:** how amplitude changes over time.
- **Spectral Plot:** how the frequency of a sound wave changes with respect to its amplitude.
- **Spectrogram:** a plot with time on the x-axis, frequency on the y-axis, and amplitude represented by the color on the plot. This allows for tracking how frequency and intensity change over time.

A Fourier transform is used to break the signals from the audio files into the amplitudes and frequencies, which are plotted in the visual representations previously listed. The prepared visual data is then fed into an ML model that can extract the features. The particulars of any model are proprietary information, but some basic analyses that we might expect include:

- **Amplitude envelope (AE):** traces the amplitude peaks in a frame of data, which correspond to distinct parts of the audio recording. The profile of audio peaks can also be used for music genre classification.
- **Short-time energy (STE):** intensity variation in a short time frame; can be used to separate voiced/unvoiced segments.
- **Root mean square energy (RMSE):** the average intensity of the signal, which can be useful for segmenting tracks into distinct sections for further analysis.
- **Zero-crossing rate (ZCR):** measures how often the signal wave crosses the horizontal axis in a frame. This can detect the presence/absence of speech, and differentiate music from speech.

2.3 Why is it Important to Understand Features of Songs?

An understanding of the features of songs and what makes them enjoyable can enable new developments in technology and business. For example, being able to quantify and analyze the features of a song can power recommendation algorithms that music services such as Spotify, Apple Music,

Pandora, etc. use to recommend songs and/or artists we may not have ordinarily chosen to listen to. Analysis of these song audio features can also enable record companies to find unsigned artists that have potential to make future hit songs. For instance, Snafu Records [11] does just this. Finally, understanding the features of songs can enable new developments in music generated using artificial intelligence ("AI-generated music"). Products such as Amper Music, OpenAI MuseNet, and others [4] use machine learning algorithms to analyze the patterns, chords, tempo, duration, and other features of songs to generate original melodies that are likely to be popular to consumers.

2.4 Previous Efforts to Understand Song Features and Popularity

Given the compelling business and technological motivations to understand what features contribute to a song's popularity, it is unsurprising that there have been many efforts in the past to investigate these relationships. For instance, the work done in [12] and [9] found that traits of popular songs had varied over time. For instance, the tempo of songs was slower in the 1980s and 1990s than the faster-paced songs of today, most popular songs use a simple time signature and hit songs are getting relatively louder. Other efforts have shown that in particular, the loudness and energy of songs went up during the so-called "Loudness Wars" of the late 1980s to early 2000s [18] showed that there has been a downward trend in the 'mood' of hit songs since the 1980s, while at the same time, the relative 'danceability' of songs has gone up, reflecting the popularity and spread of dance-based pop as opposed to a rock-and-roll type song.

3 Methodology and Design of the Project

3.1 Finding the Most Popular Songs

To start a study comparing the most popular songs, we first need a large dataset of popular songs that spans many years. The Billboard Hot 100 is the music industry standard record chart in the United States, ranking the top 100 songs based on physical and digital sales, radio play, and online streaming on a weekly basis. The "Billboard 'The Hot 100' Songs" data set from Kaggle [10] compiles these rankings for the past several decades (1958-2021) and is a good starting list for popular songs over the years.

3.2 Gathering Data About the Songs

For each song in the Billboard chart that we gather, we can collect high-level audio features on the song to compare them with one another. Spotify provides web APIs [15] that provide features about tracks derived from the analysis mentioned in section 2.2, such as:

- **Danceability:** how suitable is a song for dancing, based on tempo, rhythm stability, beat strength, and other musical elements

- **Energy:** a perceptual measure of intensity and activity (ex. Energetic tracks will feel fast/loud/noisy)
- **Instrumentalness:** how much a song relies on vocals (ex. rap relies on vocals heavily, classical music has almost no vocals)
- **Loudness:** how loud the sound is in dB, the primary psychological correlate of physical strength.

The full list of features and their definitions can be found in Table 1.

3.3 Storing the Audio Features

The song features and chart data were stored in .csv files. There were approximately 330,000 record chart listings (a listing is a song, the week of the ranking, its rank that week, how long it has been on the chart, the highest point it ever reached on the chart to that date), approximately 29,000 distinct songs, and 9,000 distinct artists. While a large amount of data to work with manually, these are manageable amounts of data to work with in memory, so I chose to store them in .csv files and conduct the analysis in Python using pandas DataFrames. Since I am more familiar with working in SQL syntax, I used the pandasql package to mimic those DataFrames being stored in a SQLite database so that I could interact with them using SQL syntax.

3.4 Obtaining Basic Descriptions of the Features

First, I wanted to plot how the features of songs had changed over time, and the distribution of each feature. Knowing how the songs changed over time would allow me to verify the observations in [9], [12], [7], and the distributions of the features would better inform me on methods to analyze the data. Finally, I generated a correlation heatmap to identify if there were relationships between different features of a song. I chose to plot the Spearman correlation, because I wanted to see if features were monotonically related, even if that relationship was not linear [16].

3.5 Grouping Songs That Become Popular

After the basic analysis above, I wanted to see if there were certain types of songs that would become popular. Without doing much analysis, it is readily apparent that the songs that become popular are not the same genre, and don't all share the same tempo, energy, etc. Simply put, there are songs that rise to the top of the charts as "rock", "hip-hop", "electronic dance music", and other types of genres. I wanted to cluster the songs that had become popular per the Billboard Top 100 dataset based on their audio features to see if there were distinct sets of songs that would become popular. For this purpose, I used K-means clustering, which is a well studied approach to finding collections of observations that share similar characteristics [5]. There had also been prior work in using K-means clustering on songs based on their attributes, from the work done in [14].

3.6 Predicting Whether a Song Will Be Popular

Finally, I wanted to examine if it was possible to predict whether a song would become popular based on its audio features. One potential measure of popularity is the Spotify meta-feature of current popularity, which is calculated to be between 0-100 for any song. The closer a song is to a popularity of 100, the more that song is played (especially recently). Another measure of popularity is how well that song performed on the charts; how high did the song actually rise on the charts, and how long was the song on the chart? I derived a measure of "Chart Performance" based on the distributions of peak rank (the highest rank the song ever achieved) and a song's maximum weeks on the chart (how long the song remained popular per the Billboard ranking). "Chart Performance" in this context would be a song that was ever ranked higher than 20 on the charts and that stayed on the chart for longer than 20 weeks. This roughly corresponds to being in the top quartile for both peak rank and weeks on the board, rounded to the nearest multiple of 10 for a more 'clean' presentation.

For both types of predictions, I used a binary logistic regression model. A logistic regression model will estimate the probability of an event based on a set of independent variables. I chose this method because it does not require a linear correlation between an attribute and the measure it is trying to predict, it allows for both discrete and continuous data to be included in the model, and because I initially wanted something that would simply classify a song as either "popular" or "not popular" and logistic regression is a good choice for this type of classification [17].

To choose the variables to include in the logistic models, I used backward stepwise elimination [13]. I started by training the models using all of the features, then looking at the p -values for each feature. The p -value indicates the likelihood the feature is significantly contributing to the prediction, so the closer the score is to 0, the more likely that feature is important in the calculation. I would remove the feature with the highest p value, train the model again, and repeat until I had a set of features with p -values below 0.001.

To evaluate the performance of the models, I used two methods. The first of which is plotting the receiver operating characteristic (ROC) curve, which shows the performance of a classification model at all classification thresholds. The ROC curve plots the *sensitivity*, or true positive rate (TPR) against the *specificity*, or false positive rate (FPR) at all decision thresholds. The decision threshold is interpreted as the probability of a classification, that if the model predicts a higher probability, we will classify as a positive. With an ROC curve, we can calculate the area under the curve (AUC), which provides an aggregate measure of the performance across all decision thresholds. The second method of evaluating the performance is to show a confusion matrix for a test dataset. In this method, the test dataset is labeled and we run

the dataset through the model with a specific decision threshold. In a binary logistic classification (like in this project) we get 4 quadrants: (true positive, predicted positive), (true positive, predicted negative), (true negative, predicted positive), (true negative, predicted negative). This is another way to visualize how often the model is predicting the true label, and if it is tending to over classify in a particular manner.

The optimal decision threshold for both models was computed using the geometric mean [3] of the sensitivity and specificity for each possible threshold, and then using the maximum of all the calculated thresholds.

4 Evaluation

4.1 Distributions of the Features

I first generated histograms of the audio features to observe their distributions. These are shown in Figure 1. Many of the features exhibited some degree of normality (ex. danceability, energy, loudness, valence, tempo). However, some features were skewed in an interesting way.

For instance, speechiness in the popular songs is mostly between 0.1-0.2, which indicates some level of spoken words (as opposed to singing or instruments) is present in popular songs. Per the Spotify API documentation, a speechiness value of 0.33 would indicate a rap song, whereas a speechiness value of 0 would be a fully instrumental song, so most songs fall somewhere in between this range. While this is an intuitive result after evaluation, it does help to set the stage of what values of speechiness to expect in these tracks.

Another interesting result is the acousticness of the songs. These values are all for the most part concentrated below 0.4 which would mean that in general, the songs on the top charts use more electronic sounds than acoustic instruments. Again, this is a somewhat intuitive result after evaluation, because there has been increasing amounts of electronic and pop songs released especially in the last couple of decades, but the plot helps to visualize the trend.

4.2 Analyzing Features Over Time

For each feature, I plotted its average value for the week, for all weeks in the dataset. These plots are in Figure 2. From these plots, we can see that some features have noticeable upward or downward trends since the start of the dataset. In particular, we can observe that danceability, energy, loudness, and speechiness, and tempo have all gone up; we can also observe that the acousticness, instrumentality, and valence have all gone down. This generally points to a trend that songs have gotten more energetic and faster paced, with a less cheerful mood and with less natural instruments. These findings line up quite well with the findings of (citation here).

4.3 Feature Correlations

To check if any features have a strong association with each other, I made a correlation heatmap, which is shown in Figure 3. The correlations that are calculated are Spearman correlations, to capture correlations that may not be linear in nature. In the plot, we can see that the strongest positive correlations ($R^2 > 0.25$) are between:

- Energy and Loudness ($R^2=0.67$)
- Valence and Danceability ($R^2=0.37$)
- Energy and Valence ($R^2=0.33$)
- Danceability and Speechiness ($R^2=0.32$)
- Speechiness and Energy ($R^2=0.31$)

We can also see the strongest negative correlations ($R^2 < -0.25$) are between:

- Energy and Acousticness ($R^2=-0.55$)
- Loudness and Acousticness ($R^2=-0.4$)
- Danceability and Acousticness ($R^2=0.25$)
- Loudness and Instrumentality ($R^2=0.32$)

From the previous time plots, we know that traits like energy, loudness, and danceability have gone up; whereas acousticness and instrumentality have gone down. The heatmap confirms this and also generally supports the sentiment in [9] that songs, in general, have gotten more loud, and energetic, exhibit more traits of dance-pop, and rely less on natural instruments.

4.4 Clustering Songs

To begin clustering songs in the dataset, I first needed to determine how many clusters I wanted to group the songs into. To do this, I used the "elbow method" heuristic, which involves running K -means clustering on the dataset for a range of values of K . For each value of K , a *distortion* score is calculated, which calculates the sum of square distances from each point to its assigned center. The distortion scores are plotted for each value of K , and the resulting plot will have an inflection point around the value of K that will be most appropriate for this dataset. The "elbow" plot I generated for this dataset is in Figure 4. The inflection point on that plot is somewhere between $K=2$ and $K=4$, so I chose a K of 4.

After running K -means clustering for the songs in the dataset, I wanted to get a profile of the songs that were in each cluster to observe common traits those songs had. I did this with two different kinds of plots, a *radar chart* to show the average of audio features that had a somewhat 'normal' distribution as per Figure 1, and a *word cloud* to see if there was a common genre amongst the cluster. From the plots, we can make the following observations about the clusters:

- **Cluster 1** is characterized generally by lower energy songs, lower danceability, lower valence, and a large number of acoustic instruments.

- **Cluster 2** is characterized generally by high energy and loudness, with a high degree of danceability and low amounts of acoustic instruments.
- **Cluster 3** seems to generally be a higher valence (i.e. more cheerful) version of Cluster 2.
- Similarly, **Cluster 4** is very generally a higher valence version of Cluster 1.

The plots to characterize each of the clusters can be found in Figures 5 through 8 below.

There seems to be a commonality with the songs in clusters 1 and 4, and clusters 2 and 3. Another observation from the clusters is the number of songs in each. Clusters 1 and 4 have 4500 and 5500 songs each, respectively. Clusters 2 and 3 have 8000 and 10000 songs each, respectively. This means there are more 'high-energy, high-danceability, high-loudness, low acoustic songs in the dataset, which lines up with observations from past sections and with the previously cited related work [9].

4.5 Predicting a Song's Popularity

As stated in Section (add section), I used a logistic regression model to predict whether a song would be popular, based on both the popularity score as calculated by Spotify and the "Chart Performance" metric detailed in section (add section). I used the tools from `scikit-learn` to calculate the models.

For the current popularity prediction model, the coefficients are in Figure 10. As shown in Figure 1, the distribution of the current popularity of the songs resembles a somewhat normal distribution, so I decided to classify songs as "popular" or "not popular" based on if they were above or below the median current popularity. The performance of this model is shown in Figure 13. The model turned out to be reasonably ok at predicting whether a song would be popular or not, with an AUC=0.746. Looking at the confusion matrix over the test set, the song correctly classified "popular" songs as "popular" and "not popular" songs as "not popular" the majority of the time. The model has a bias toward predicting true negatives, as the ratio of classification of "not popular" songs was better here.

For these models, the variables with the largest coefficients are the ones where an increase or decrease in them would cause the largest change in the predicted popularity classification. For the current popularity prediction model, the largest 3 were **danceability**, **speechiness**, and **energy**. This would suggest that these are the key features that would predict whether a song would be popular or not today. This lines up well with the findings in the feature time plots of Figure 2 and the previous work in this field, both of which found that these are traits that have been increasing in popular music over time.

For the chart performance prediction model, the coefficients are in Figure 11. As shown in Figure 9, the peak ranks of songs appear to be mostly uniform, except for a spike in

the number of songs that reached the top 5 on the chart. The distribution of "weeks on the board" for each song is skewed right, as most songs stay on the chart for less than 20 weeks. Using these distributions, I wanted to isolate songs that were top performers with regard to both peak rank and weeks on the board, and this is why I classified a "top chart performer" as a song that achieved a peak rank of 1-20 and stayed on the board for over 20 weeks. This roughly corresponds to being in the upper quartile of both distributions. The performance of this model is shown in Figure 11. The model turned out to be slightly less capable of predicting whether a song would be a top chart performer, with an AUC=0.676. Looking at the confusion matrix over the test set, the song correctly classified "not top performers" songs as "not top performers" the majority of the time. However, it classified "top chart performers" songs as "not top performers" the majority of the time (i.e. the false negative rate was high).

For the chart performance prediction model, the largest 3 coefficients were **danceability**, **speechiness**, and **acousticness**. The top two of danceability and energy line up with the findings from the popularity prediction model. However, in this model, acousticness was an important attribute. This may be because although popular songs today use far more electronic instruments than acoustic ones, the chart ranking is how that song performed in the past. We also know that songs used to be more acoustic in the past (Figure 2) so it is likely that the model is being influenced by songs that were top chart performers before the decline in acousticness.

5 Conclusions and Future Work

It seems we can reasonably predict if a song will be popular today or not based on its high-level extracted features, and the key features that will predict a song's popularity are: danceability, speechiness, and energy. It also seems that these are traits that have gone up over time since the beginning of the dataset in 1958. We can also see in the clustering that although some of the clusters appear to overlap each other, some of the most prominent genres of songs that have been in the Billboard Top 100 are:

- Loud, Acoustic songs: country, rock-and-roll
- Loud, Energetic, Danceable, with low mood: hip-hop, rap, electronic pop
- Loud, Energetic, Danceable, with high mood: hip-hop, new wave, and classic rock
- Loud, Acoustic, Danceable, with high mood: soft rock, classic rock, folk music

Based on those findings, we would assume that if a song does not fit into one of those profiles, it is unlikely to reach a high position on the charts or be a popular song today.

To enhance the work in the future, we could add some elements to the investigation. The first of which is trying to implement the feature extraction process detailed in section 2.2 to see if we could replicate the characterization of songs

as "loud", or "energetic", and so on. This would give a deeper understanding of how the frequency and amplitude of the sound waves in the songs contribute to a song's popularity. We could also examine top charts in other countries to examine how tastes in music between countries might differ. There is less readily available information on what songs are chart-toppers in other countries (for instance, the Billboard Global 200 has only been collecting chart data since 2019) but more investigation could be done here. We could also try different prediction models (for example, multiple polynomial regression to predict the actual popularity score). More investigation into the types of correlations between each audio feature and the popularity would need to be done though, as multiple regression comes with new assumptions about the dataset that must be met. Finally, we could do a more literature review to examine why a particular sound signature would fit into the patterns that humans enjoy, and more deeply examine the neurological and psychological aspects of why certain songs become popular.

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A Audio Feature Definitions and Figures

Feature	Definition
Acousticness	A confidence measure from 0.0 to 1.0 of whether the track is acoustic. 1.0 represents high confidence the track is acoustic.
Danceability	Danceability describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity. A value of 0.0 is least danceable and 1.0 is most danceable.
Energy	Energy is a measure from 0.0 to 1.0 and represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. For example, death metal has high energy, while a Bach prelude scores low on the scale. Perceptual features contributing to this attribute include dynamic range, perceived loudness, timbre, onset rate, and general entropy.
Instrumentalness	Predicts whether a track contains no vocals. "Ooh" and "aah" sounds are treated as instrumental in this context. Rap or spoken word tracks are clearly "vocal". The closer the instrumentalness value is to 1.0, the greater likelihood the track contains no vocal content. Values above 0.5 are intended to represent instrumental tracks, but confidence is higher as the value approaches 1.0.
Key	The key the track is in. Integers map to pitches using standard Pitch Class notation. E.g. 0 = C, 1 = C-sharp/D-flat, 2 = D, and so on. If no key was detected, the value is -1.
Liveness	Detects the presence of an audience in the recording. Higher liveness values represent an increased probability that the track was performed live. A value above 0.8 provides strong likelihood that the track is live.
Loudness	The overall loudness of a track in decibels (dB). Loudness values are averaged across the entire track and are useful for comparing relative loudness of tracks. Loudness is the quality of a sound that is the primary psychological correlate of physical strength (amplitude). Values typically range between -60 and 0 db.
Mode	Mode indicates the modality (major or minor) of a track, the type of scale from which its melodic content is derived. Major is represented by 1 and minor is 0.
Speechiness	Speechiness detects the presence of spoken words in a track. The more exclusively speech-like the recording (e.g. talk show, audio book, poetry), the closer to 1.0 the attribute value. Values above 0.66 describe tracks that are probably made entirely of spoken words. Values between 0.33 and 0.66 describe tracks that may contain both music and speech, either in sections or layered, including such cases as rap music. Values below 0.33 most likely represent music and other non-speech-like tracks.
Tempo	The overall estimated tempo of a track in beats per minute (BPM). In musical terminology, tempo is the speed or pace of a given piece and derives directly from the average beat duration.
Time Signature	An estimated time signature. The time signature (meter) is a notational convention to specify how many beats are in each bar (or measure). The time signature ranges from 3 to 7 indicating time signatures of "3/4", to "7/4".
Valence	A measure from 0.0 to 1.0 describing the musical positiveness conveyed by a track. Tracks with high valence sound more positive (e.g. happy, cheerful, euphoric), while tracks with low valence sound more negative (e.g. sad, depressed, angry).

Table 1. Definitions of song audio features referenced throughout the paper. Definitions are provided by Spotify.^[15]

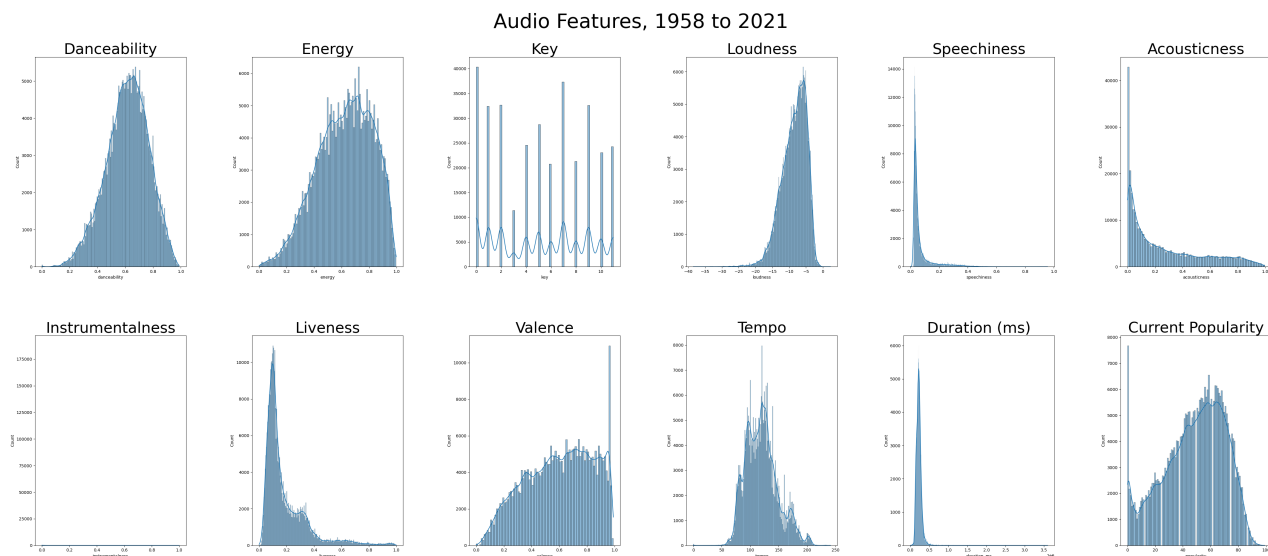


Figure 1. Audio Features of the Billboard Top 100 songs since 1958, plotted as histograms.

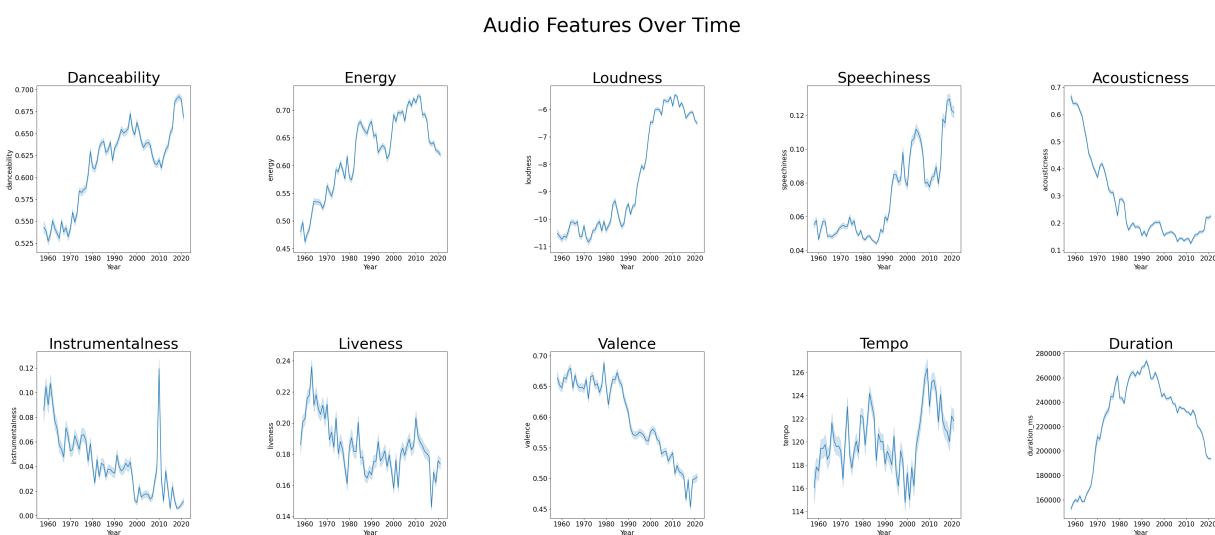


Figure 2. Audio Features of the Billboard Top 100 songs since 1958, plotted over time.

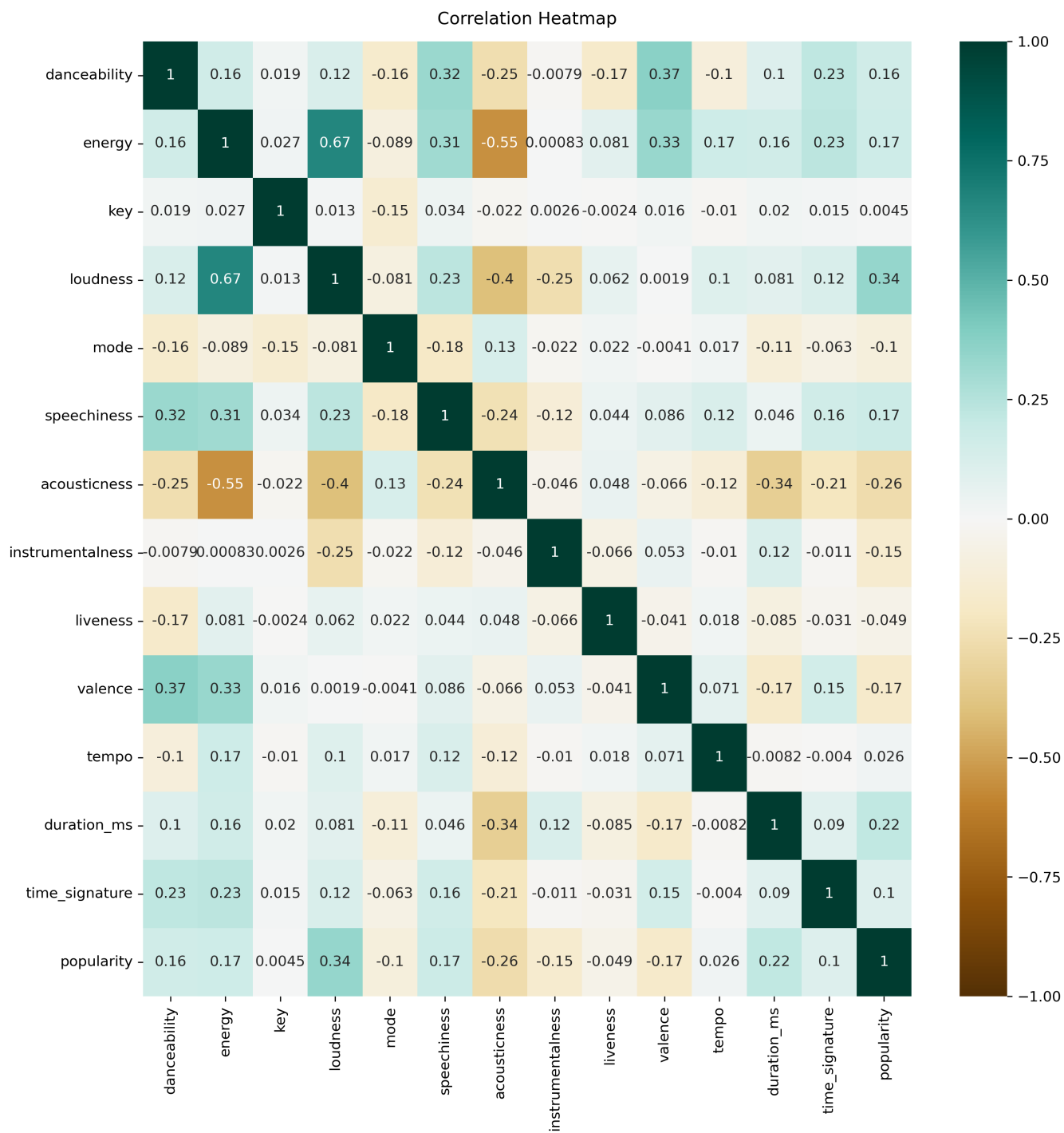


Figure 3. Correlation Heatmap of Audio Features

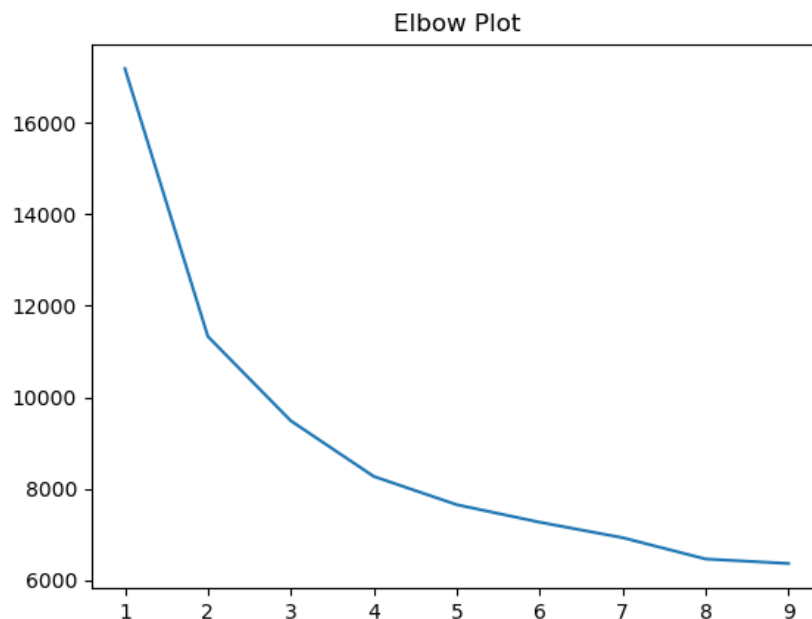


Figure 4. "Elbow" Plot to determine the ideal number of clusters for K -means clustering.

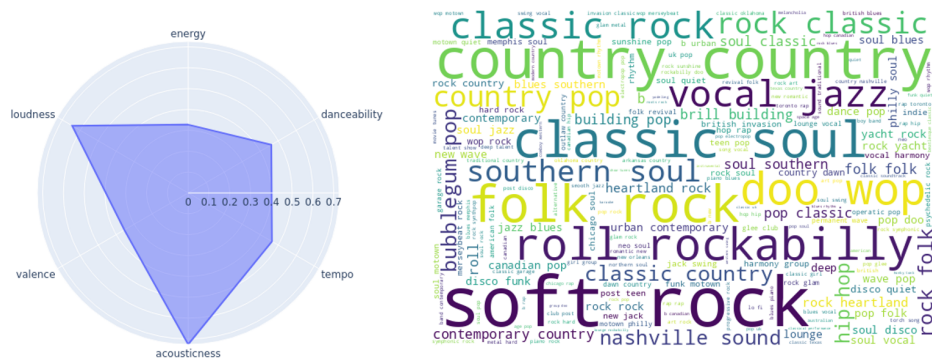


Figure 5. Characterizing the first cluster produced by K -means clustering for $K=4$.

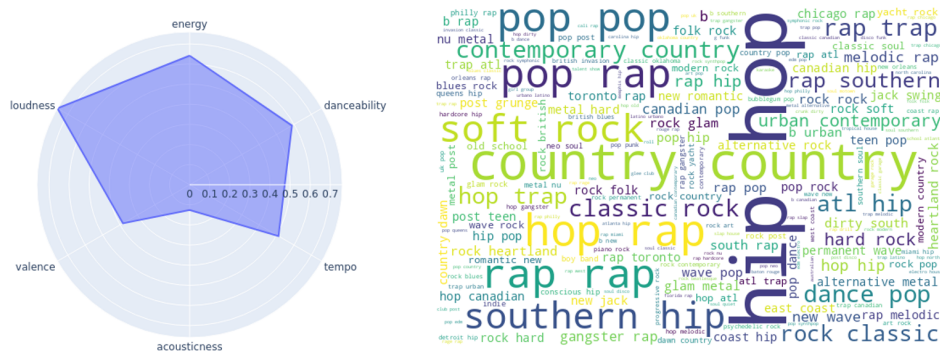


Figure 6. Characterizing the second cluster produced by K -means clustering for $K=4$.

Logit Regression Results						
=====						
Dep. Variable:	is_popular	No. Observations:	21363			
Model:	Logit	Df Residuals:	21351			
Method:	MLE	Df Model:	11			
Date:	Tue, 02 May 2023	Pseudo R-squ.:	0.1355			
Time:	01:19:19	Log-Likelihood:	-12542.			
converged:	True	LL-Null:	-14508.			
Covariance Type:	nonrobust	LLR p-value:	0.000			
=====						
	coef	std err	z	P> z	[0.025	0.975]
danceability	2.0577	0.123	16.790	0.000	1.817	2.298
energy	-0.6812	0.131	-5.218	0.000	-0.937	-0.425
loudness	0.1474	0.006	23.768	0.000	0.135	0.160
mode	-0.1419	0.034	-4.150	0.000	-0.209	-0.075
speechiness	2.0511	0.192	10.671	0.000	1.674	2.428
acousticness	-0.9769	0.068	-14.273	0.000	-1.111	-0.843
instrumentalness	-1.1786	0.110	-10.688	0.000	-1.395	-0.963
liveness	-0.5928	0.100	-5.911	0.000	-0.789	-0.396
valence	-1.9753	0.082	-24.070	0.000	-2.136	-1.814
tempo	0.0024	0.001	4.504	0.000	0.001	0.003
duration_ms	1.238e-06	2.4e-07	5.154	0.000	7.67e-07	1.71e-06
time_signature	0.2800	0.036	7.758	0.000	0.209	0.351
=====						

Figure 10. Logistic regression model for predicting if a song will be "popular" (above the median current popularity of all songs on the charts) or "not popular" (below the median current popularity of all songs on the charts).

Logit Regression Results						
=====						
Dep. Variable:	chart	No. Observations:	21363			
Model:	Logit	Df Residuals:	21351			
Method:	MLE	Df Model:	11			
Date:	Tue, 02 May 2023	Pseudo R-squ.:	0.05941			
Time:	01:18:35	Log-Likelihood:	-7390.6			
converged:	True	LL-Null:	-7857.4			
Covariance Type:	nonrobust	LLR p-value:	3.558e-193			
=====						
	coef	std err	z	P> z	[0.025	0.975]

danceability	1.6953	0.170	9.996	0.000	1.363	2.028
energy	-1.9040	0.183	-10.377	0.000	-2.264	-1.544
loudness	0.1122	0.009	12.674	0.000	0.095	0.130
mode	-0.2372	0.046	-5.170	0.000	-0.327	-0.147
speechiness	-0.8532	0.266	-3.202	0.001	-1.375	-0.331
acousticness	-1.2811	0.104	-12.323	0.000	-1.485	-1.077
instrumentalness	-0.4404	0.156	-2.827	0.005	-0.746	-0.135
liveness	-0.6880	0.156	-4.416	0.000	-0.993	-0.383
valence	-0.5099	0.110	-4.649	0.000	-0.725	-0.295
tempo	-0.0021	0.001	-2.644	0.008	-0.004	-0.001
duration_ms	4.052e-06	3.05e-07	13.273	0.000	3.45e-06	4.65e-06
time_signature	-0.1482	0.050	-2.937	0.003	-0.247	-0.049
=====						

Figure 11. Logistic regression model for predicting if a song will be a "good chart performer" (peak rank of at least 20 and stayed on the charts for at least 20 weeks) or "not a good chart performer" (below the median current popularity of all songs on the charts).

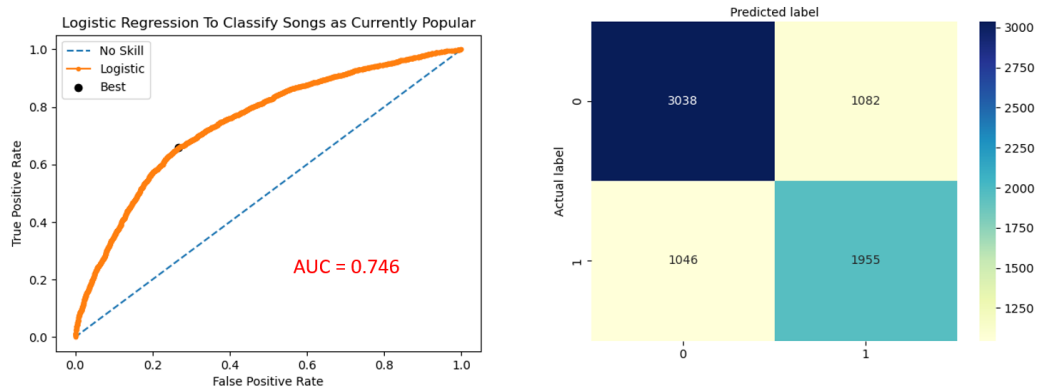


Figure 12. ROC curve for logistic regression model for predicting if a song will be "popular" (above the median current popularity of all songs on the charts) or "not popular" (below the median current popularity of all songs on the charts).

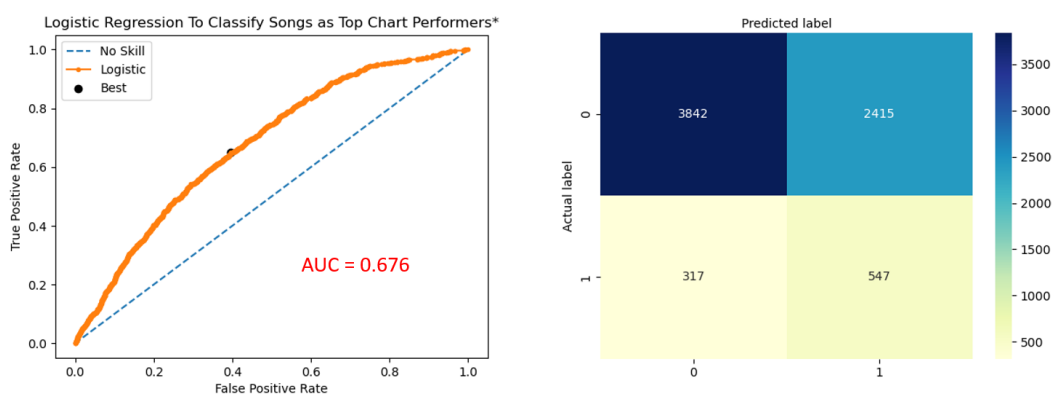


Figure 13. ROC curve for logistic regression model for predicting if a song will be a "good chart performer" (peak rank of at least 20 and stayed on the charts for at least 20 weeks) or "not a good chart performer" (below the median current popularity of all songs on the charts).